Skills Summary

Polynomial Long Division

Use this reference while completing your worksheet or when studying.

Skill 1 — Division That Comes Out Even

The four-step loop: divide the leading terms, multiply the divisor by that result, subtract the whole line, bring down the next term. Repeat until what is left has lower degree than the divisor.

Check: quotient $\times$ divisor should rebuild the dividend.

Example: Divide $\dfrac{x^2+9x+20}{x+4}$

Step 1. Both leading terms are x-terms, so the first piece of the quotient is whatever multiplies x into x^2 — start the loop there.

Step 2. $x(x+4) = x^2+4x$; subtracting leaves $5x+20$. Then $5(x+4) = 5x+20$, and the subtraction leaves nothing.

$\Rightarrow \quad \boldsymbol{x+5}$

Try it: Divide $\dfrac{x^2-x-12}{x-4}$

check: $\boldsymbol{x+3}$

Skill 2 — Quotient Plus a Remainder

Stop rule: stop when the leftover has lower degree than the divisor. That leftover is the remainder.

Report it as quotient + remainder over divisor. A negative remainder is reported with a minus sign in front of the fraction, not hidden.

Example: Divide $\dfrac{2x^2+3x-1}{x+1}$

Step 1. The divisor has degree 1, so the loop runs until the leftover is a constant — that constant is the remainder, not an error.

Step 2. $2x(x+1) = 2x^2+2x$, leaving $x-1$. Then $1(x+1) = x+1$, leaving -2, which has degree 0: stop.

$\Rightarrow \quad \boldsymbol{2x+1-\dfrac{2}{x+1}}$

Try it: Divide $\dfrac{3x^2+2x+7}{x+2}$

check: $\boldsymbol{3x-4+\dfrac{15}{x+2}}$

Skill 3 — Zero Placeholders for Missing Powers

Rule: before dividing, rewrite the dividend with every descending power present, using a 0 coefficient where a power is missing — for example $x^3 - 8$ becomes $x^3 + 0x^2 + 0x - 8$.

Without the placeholders the columns slide and terms get subtracted from the wrong partners.

Example: Divide $\dfrac{x^3 + 27}{x + 3}$

Step 1. Two powers are missing from the dividend, so write them in with zero coefficients before starting the loop — that is the whole difficulty here.

Step 2. Dividing $x^3 + 0x^2 + 0x + 27$ gives x^2, then $-3x$, then 9, and the final subtraction leaves nothing.

$\Rightarrow$ $\boldsymbol{x^2 - 3x + 9}$

Try it: Divide $\dfrac{x^3 - 1}{x - 1}$

check: $\boldsymbol{x^2 + x + 1}$

Skill 4 — The Remainder Theorem Shortcut

Remainder theorem: dividing a polynomial P by $x - a$ leaves the remainder $P(a)$. So the remainder can be found by one substitution, with no division at all.

Factor theorem: $P(a) = 0$ exactly when $x - a$ is a factor. For a divisor $x + 3$, the value to substitute is $a = -3$.

Example: Find the remainder when $x^3 - 4x^2 + x + 6$ is divided by $x - 1$.

Step 1. Only the remainder is wanted, not the quotient, so the theorem replaces the whole division with one substitution at $a = 1$.

Step 2. $1 - 4 + 1 + 6$, which is the same number the last subtraction of a full long division would leave.

$\Rightarrow$ **4**

Try it: Find the remainder when $2x^3 + x - 5$ is divided by $x + 2$.

check: $\boldsymbol{-23}$